Deterministic Optimization

Discrete Optimization:

Modeling w/ binary variables

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Modeling Exercises 1

Modeling Exercises 1

Learning Objectives

- Observe modeling a facility location problem
- Observe modeling a generation scheduling problem

A Facility Location Problem

- We need to pick sites to locate distribution centers (DC) from a set of n potential sites
- Let f_j and v_j denote the fixed and variable (per-unit capacity) cost of locating at site j, respectively
- The DCs will serve m retail locations and the demand of retail i is d_i
- The per-unit-cost of transporting from DC location j to retail location i is c_{ij}
- Decide where to locate the DCs, how much capacity to install, and how to match DC capacity to retail demand at minimum total cost

A Facility Location Problem

Variables:
y_j: Binary variable indicating whether location j is chosen
x_j: Continuous variable indicating the capacity of DC located at j
z_{ij}: Units shipped from DC j to retail location i

$$
\begin{aligned}
\min \quad & \sum_{j=1}^{n}(f_j y_j + v_j x_j) + \sum_{i=1}^{m}\sum_{j=1}^{n} c_{ij} z_{ij} \\
\text{s.t.} \quad & \sum_{j=1}^{n} z_{ij} = d_i && \forall\ i = 1, \ldots, m \\
& \sum_{i=1}^{m} z_{ij} \leq x_j && \forall\ j = 1, \ldots, n \\
& 0 \leq x_j \leq (\textstyle\sum_{i=1}^{m} d_i) y_j && \forall\ j = 1, \ldots, n \\
& 0 \leq z_{ij} \leq d_i y_j && \forall\ i = 1, \ldots, m,\ j = 1, \ldots, n \\
& y_j \in \{0, 1\} && \forall\ j = 1, \ldots, n
\end{aligned}
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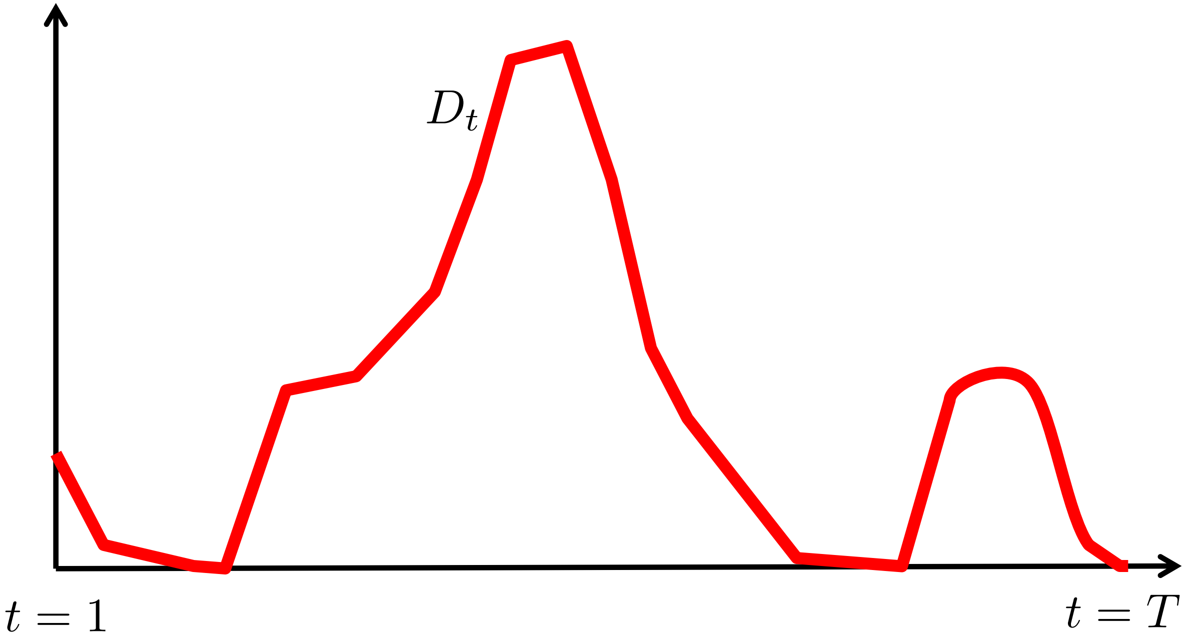
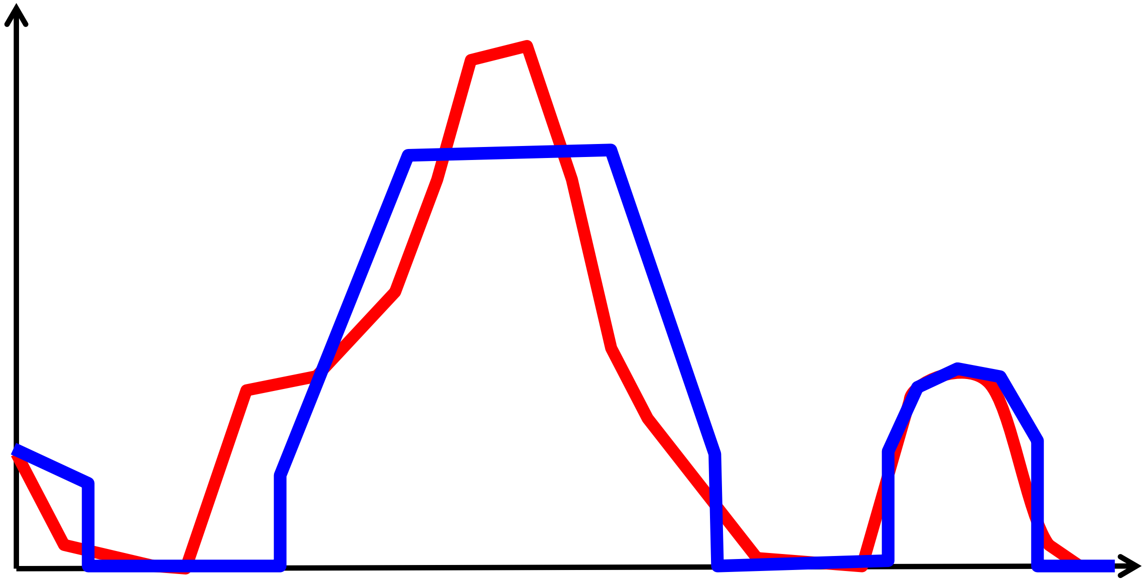
A Generator Scheduling Problem

Schedule generator operation to meet load over a time horizon



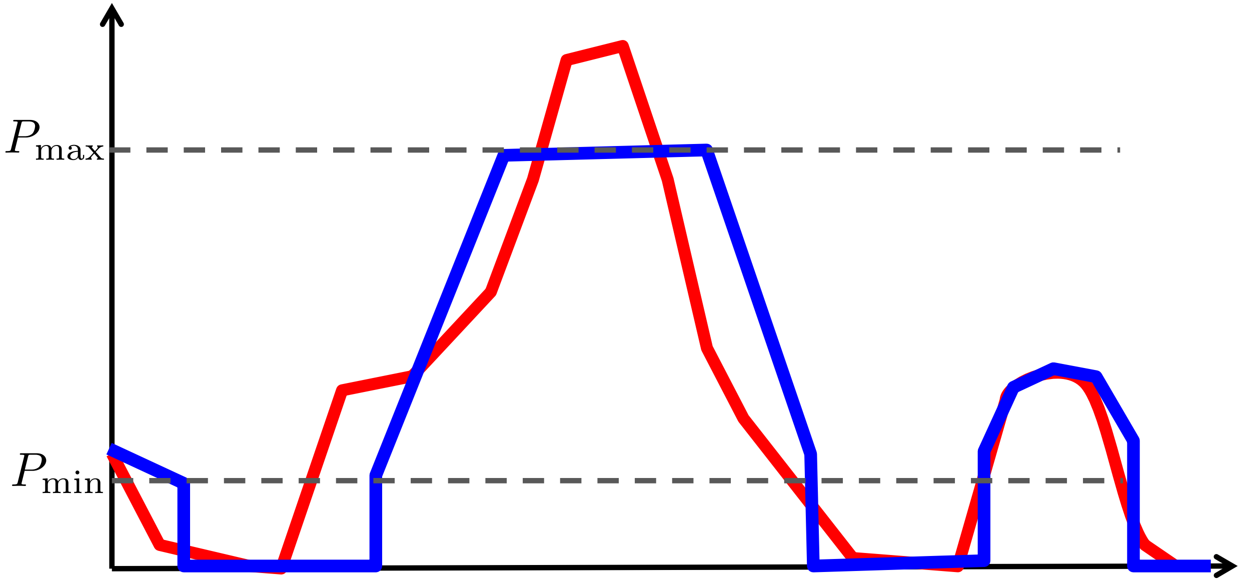
A Generator Scheduling Problem

Generation schedule must satisfy some operational constraints



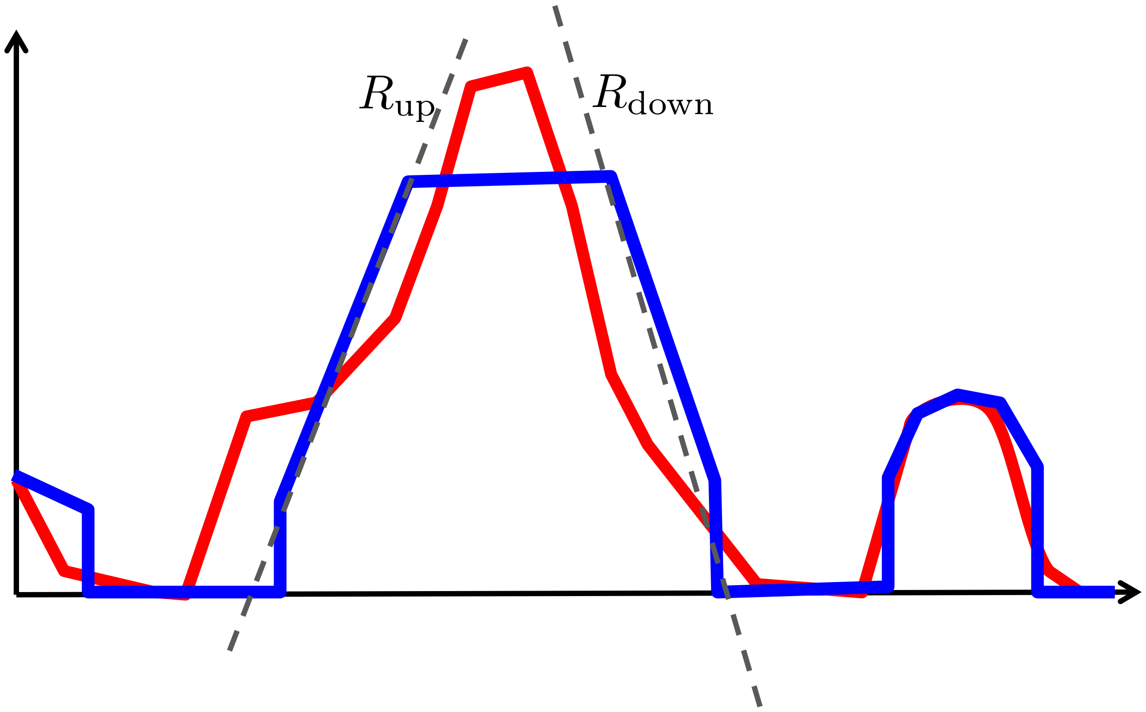
A Generator Scheduling Problem

Min/Max Production Limits: If generator is on it must produce within given min/max limits



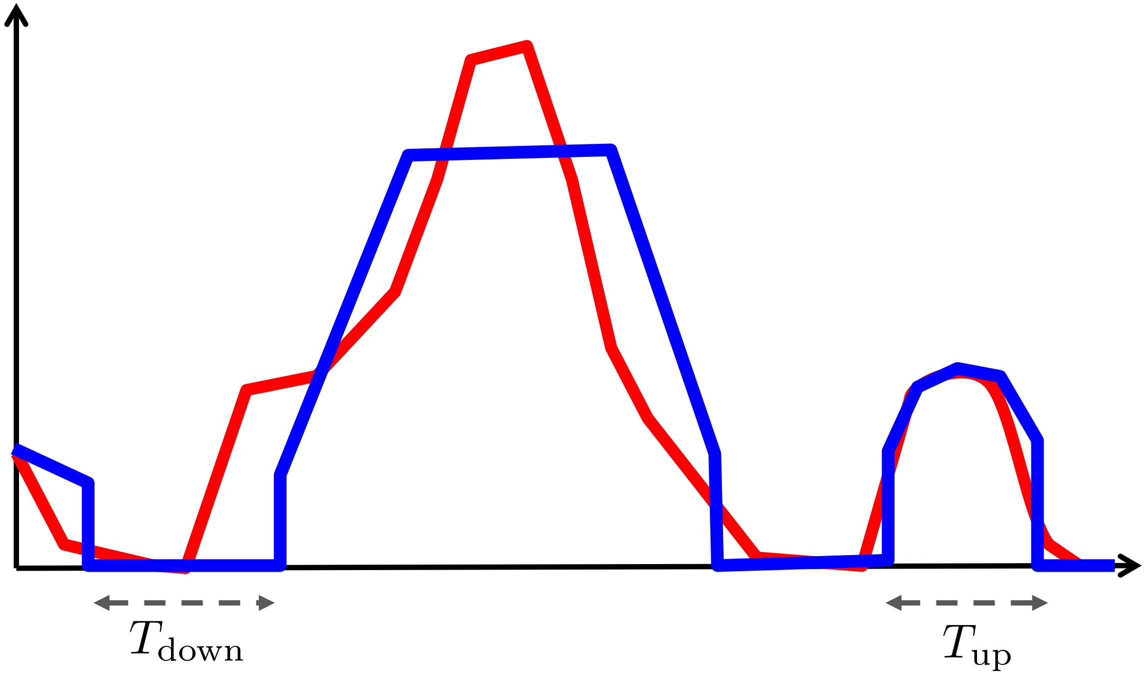
A Generator Scheduling Problem

Ramp up/down limits: Production cannot increase/decrease faster than prescribed limits



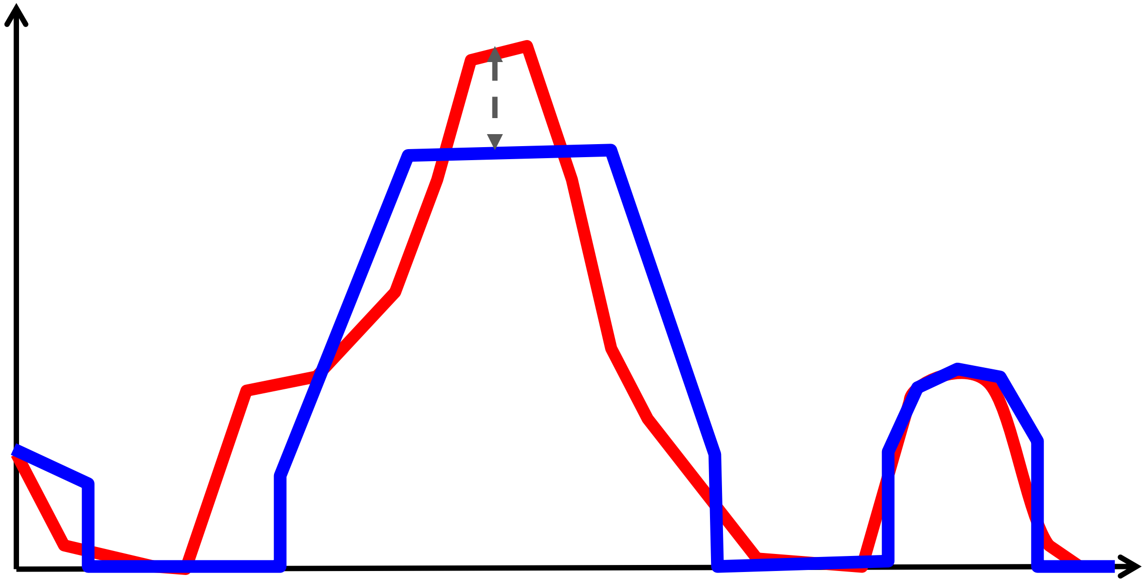
A Generator Scheduling Problem

Min up/down time limits: If a generator is turned off, it must be off for a certain minimum time. Similarly, once it is turned, it must stay one for a certain amount of time



A Generator Scheduling Problem

Total cost = generation cost + cost of unmet load (spot price)



C = Per-unit generation cost $\qquad S_t$ = Spot cost in period t

A Generator Scheduling Problem

Variables:
x_t: Generation in period t ($t = 1, \ldots, T$)
z_t: Power from spot market in period t
y_t: Status (on/off) of generator in period t

$$
\begin{array}{lll}
\min & \displaystyle\sum_{t=1}^{T} (C x_t + S_t z_t) & \\
\text{s.t.} & x_t + z_t = D_t & t = 1, \ldots, T \\
& P_{\min} y_t \le x_t \le P_{\max} y_t & t = 1, \ldots, T \\
& x_t - x_{t-1} \le R_{\text{up}} & t = 2, \ldots, T \\
& x_{t-1} - x_t \le R_{\text{down}} & t = 2, \ldots, T \\
& x_t, z_t \ge 0,\ y_t \in \{0, 1\} & t = 1, \ldots, T \\
& + \text{ min up/down constraints} &
\end{array}
$$

Min Up Constraints

$$y_\tau \geq y_t - y_{t-1} \;\forall\; \tau = t+1, \ldots, \min\{t + T_{\text{up}}, T\}, \;\forall\; t = 1, \ldots, T-1$$

Example: Suppose $T = 6, T_{\text{up}} = 3$ and $y_0 = 0$

$$\begin{aligned}
t = 1 &: \; y_2 \geq y_1, \; y_3 \geq y_1 \\
t = 2 &: \; y_3 \geq y_2 - y_1, \; y_4 \geq y_2 - y_1 \\
t = 3 &: \; y_4 \geq y_3 - y_2, \; y_5 \geq y_3 - y_2 \\
t = 4 &: \; y_5 \geq y_4 - y_3, \; y_6 \geq y_4 - y_3 \\
t = 5 &: \; y_6 \geq y_5 - y_4
\end{aligned}$$

Min Down Constraints

$$y_\tau \leq y_t - y_{t-1} + 1 \; \forall \; \tau = t+1, \ldots, \min\{t + T_{\text{down}}, T\}, \; \forall \; t = 1, \ldots, T-1$$

Example: Suppose $T = 6, T_{\text{down}} = 3$ and $y_0 = 0$

$$\begin{aligned}
t = 1: &\;\; y_2 \leq y_1 + 1, \;\; y_3 \leq y_1 + 1 \\
t = 2: &\;\; y_3 \leq y_2 - y_1 + 1, \;\; y_4 \leq y_2 - y_1 + 1 \\
t = 3: &\;\; y_4 \leq y_3 - y_2 + 1, \;\; y_5 \leq y_3 - y_2 + 1 \\
t = 4: &\;\; y_5 \leq y_4 - y_3 + 1, \;\; y_6 \leq y_4 - y_3 + 1 \\
t = 5: &\;\; y_6 \leq y_5 - y_4 + 1
\end{aligned}$$

Summary

- We observed some examples of modeling with integer variables